Nursing Outlook 73 (2025) 102400

Contents lists available at ScienceDirect

Nursing Outlook

journal homepage: www.nursingoutlook.org

Leader perspectives on the "experience-complexity gap": Recommendations to fortify the nursing workforce

Aditi D. Rao, PhD, RN [a,b,*], Evan P. Anderson, MSN, RN [b], Beth A. Smith, MSN, RN, NEA-BC [a], Matthew McHugh, PhD, RN, FAAN [c,d], Regina S. Cunningham, PhD, RN, FAAN [a,b]

[a] Penn Medicine, Hospital of the University of Pennsylvania, Philadelphia, PA
[b] Department of Biobehavioral Sciences, University of Pennsylvania School of Nursing, Philadelphia, PA
[c] Center for Health Outcomes and Policy Research, University of Pennsylvania School of Nursing, Philadelphia, PA
[d] Leonard Davis Institute of Health Economics, University of Pennsylvania, Philadelphia, PA

ARTICLE INFO

Article history:
Received 6 December 2024
Received in revised form 5 March 2025
Accepted 23 March 2025
Available online xxxx

Keywords:
Nursing
Workforce
Education
Competency
Transition
Retention
Quality
Safety

ABSTRACT

Background: Reported workforce shifts, characterized as the "experience-complexity gap," threaten nurses' ability to deliver safe, high-quality care. Addressing this threat requires a robust understanding of the current state, which leaders in practice settings are well-positioned to elucidate.
Purpose: Describe practice leaders' perceptions of (a) new-to-practice nurses' (NTPNs) ability to practice competently, (b) the current composition of nursing teams, and (c) the impact of team composition on quality and safety, experienced nurses, and leaders.
Methods: A web-based survey was administered to practice leaders across the country. Responses were analyzed using descriptive statistics and conventional content analysis.
Discussion: Approximately 45% of respondents believe NTPNs practice competently. Specific skills gaps, measures to support NTPNs, the strain placed on experienced nurses to maintain the standard of care, generational conflict, and perceived risks to quality and safety are described.
Conclusion: These findings should inform academic and practice leaders' partnered efforts to implement competency-based education.
© 2025 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Introduction

Prior to the pandemic, a handful of authors described the emergence of a nursing "experience-complexity gap" as patients' clinical complexity was increasing, the proportion of experienced nurses was decreasing (Advisory Board, 2019; Hill, 2010). In the aftermath of the pandemic, nurses' departure from the workforce accelerated (Martin et al., 2023), further exacerbating the experience-complexity gap and, likely, disrupting the communities of practice that historically helped to cultivate nursing expertise.

Communities of practice are collections of individuals that produce a shared practice by engaging in a process of collective learning around performing a set of joint activities (Wenger, 1998). In nursing, these communities exist in the practice setting where new-to-practice nurses (NTPNs) develop competence by transferring knowledge and skills learned in the classroom to real-world patient care (Baartman & De Bruijn, 2011). NTPNs begin gaining experience as students, learning from course instructors. As they transition to practice, their development continues as employees, learning from experienced nurses. This experiential learning is essential to grow from novice to expert (Benner, 1984; Mchugh & Lake, 2011) and, eventually, facilitate others' learning within the community of practice (Billett, 2004; Tynjälä, 2008). See Figure 1.

With fewer experienced nurses, organizations have lost both their expertise in patient care and their influence on NTPNs. Compounding diminished expertise, in recent years, a decline in initial competency (Kavanagh & Sharpnack, 2021; Kavanagh & Szweda, 2017) and a longer time to demonstrate competency among new graduate registered nurses (Advisory Board, 2019) have been noted. These workforce shifts threaten nurses' ability to deliver high-quality, safe care (Hill, 2010); addressing this threat requires a robust understanding of the current state (Auerbach et al., 2024).

This paper reports practice leaders' (e.g., nurse managers, directors, educators, senior and leaders) perceptions of (a) NTPNs' (i.e., nurses with less than 18 months experience) ability to practice competently, (b) the composition of nursing teams (i.e., the proportion of novice to experienced nurses) forming communities of practice, and (c) the impact of nursing team composition on quality

* Corresponding author: A.D. Rao, 5807 Overlook Dr., Austin, TX 78731.
E-mail address: aditirao@upenn.edu (A.D. Rao).

https://doi.org/10.1016/j.outlook.2025.102400
0029-6554/© 2025 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

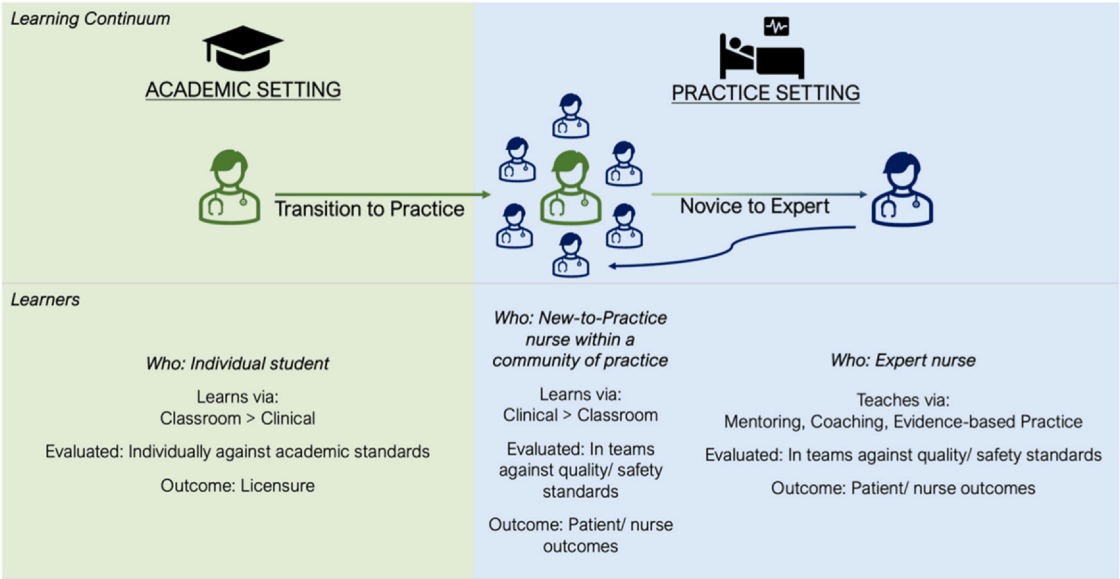

Figure 1. Nurse learners' process of collective learning along the academic-to-practice learning continuum.

and safety, experienced nurses, and nurse leaders. Amidst a national effort to implement competency-based education (CBE) as a means to strengthen nursing practice, understanding current learning processes is critical. Practice leaders are important key informants given their vantage point within the clinical learning environment. Their perspectives can help practice, academic, and policy leaders evaluate current learning processes and strengthen CBE implementation efforts.

Methods

We conducted a web-based survey to elicit perspectives from practice leaders across the country. A purposive sampling strategy was used. Nurses self-identifying as formal (i.e., nurse managers, directors, educators, clinical nurse specialists, and chief nursing officers) or informal (e.g., charge nurses) practice leaders were included; non-nurses or those who did not consent were excluded. To access potential respondents, we shared survey materials with leaders at various healthcare (e.g., health systems) and professional organizations (i.e., National Association of Clinical Nurse Specialists, Association for Nursing Professional Development, and American Organization of Nurse Leaders, Vizient) to distribute via organizational listservs. The University of Pennsylvania institutional review board approved this study as an exempt protocol.

Survey Instrument

We constructed the survey instrument using AACN's *The Essentials: Entry-Level Professional Nursing Education Competencies* (2021). We evaluated, but did not use, existing validated surveys because, although some focused on related concepts, they did not sufficiently reflect the study aims; they were not validated for use with practice leaders and did not specifically address the contemporary practice standards outlined by AACN or the element of communities of practice we aimed to explore.

Our survey instrument was designed with three sections. The first assessed the extent to which respondents agreed NTPNs "are entering the work environment prepared to" practice competently, defined using eight AACN entry-level competencies. These eight competencies were selected because they articulate key elements of how NTPNs are expected to provide direct patient care in partnership with an interprofessional team (see Table 2). The survey statements in this section were derived from competencies and subcompetencies within the Person-Centered Care, Interprofessional Partnerships, and Professionalism domains (AACN, 2021). Additionally, with each survey statement, respondents were given the opportunity to open-endedly, "describe gaps you are observing, if any." The second section explored respondents' perceptions of the composition of their nursing teams; the third assessed the extent to which respondents believe team composition impacts quality and safety, experienced nurses, and nurse leaders (see Table 3). With each question in the third section, respondents were given the opportunity to open-endedly, "describe the impact, if any." Written informed consent, describing data security and measures taken to anonymize responses, was obtained prior to releasing survey questions.

Analysis

We utilized descriptive statistics and ANOVA to analyze quantitative responses and conventional content analysis (Hsieh & Shannon, 2005), using an inductive and stepwise approach, to analyze open-ended responses. First, an experienced researcher (A.R.) developed a preliminary codebook by thoroughly reading a subset of the data and assigning first-level codes to text segments that reflected recurring ideas, then grouping codes together to form categories, and comparing all codes and categories to develop themes grounded in the responses provided. Next, a second researcher (E.A.) reviewed the preliminary codebook and applied it to a randomly selected subset of the data. The two researchers compared their codes and discussed discrepancies to refine the codebook and reach consensus on the definition of each theme. Throughout the analysis of the remaining responses, the codebook was revised. Iterative coding continued until no new themes emerged.

Several measures were taken to enhance trustworthiness. To enhance credibility, we engaged in dialog to ensure the themes reflected the content of the responses and practiced reflexivity to minimize the risk of researcher bias influencing the findings. We also reviewed themes with the larger research team, who have varying roles and perspectives, to challenge assumptions in our interpretations. To ensure dependability, we documented our analytic plan, including decisions made to refine the codebook. To ensure confirmability, we include specific examples of raw data to illustrate the themes. To ensure transferability, we include respondents' roles

Table 1
Respondent Demographics

Respondent Demographics	
Demographics (N = 396)	Mean (SD) or n (%)
Age (years)	46.2 (11.2)
Years of nursing experience	21.8 (11.4)
Years at current organization	13.7 (10.5)
Role	
Clinical nurse	38 (9.5%)
Nurse manager	127 (32.1%)
Nurse educator	82 (20.7%)
Clinical nurse specialist	56 (14.1%)
Nursing supervisor	11 (2.8%)
Director of nursing	50 (12.6%)
Chief nursing officer	11 (2.8%)
Other	21 (5.3%)
Care setting	
Ambulatory practice	24 (6.1%)
Health system	45 (11.4%)
Home health	8 (2%)
Hospital	309 (78%)
LTC/SNF	4 (1%)
Other	5 (1.3%)
Missing	1 (0.3%)
Organization volume	
Number of beds	546.3 (450.7)
Average monthly patient visits	23,265.2 (10,3944.6)
Region	
Midwest	44 (11.1%)
Northeast	271 (68.4%)
South	62 (15.7%)
West	16 (4.0%)
Missing	3 (0.8%)

Note. LTC/SNF, long-term care or skilled nursing facility.

with examples of the raw data to better contextualize each response (Elo & Kyngäs, 2008; Graneheim & Lundman, 2004; Korstjens & Moser, 2018; Lincoln & Guba, 1985).

Findings

Three hundred ninety-six respondents were included in the final dataset. Respondents who did not consent (n = 5) or those who answered only demographic questions (n = 166) were excluded. Respondents were, on average, 46 years old, with 22 years of nursing experience, and 14 years of organizational tenure. Most were frontline leaders (i.e., nurse managers = 32%, nurse educators = 21%, and clinical nurse specialists = 14%) working in hospitals (78%). Those listed as "other" typically included nurses in quality and safety, implementation, and/or informatics roles who were not specifically listed as directors or managers. Table 1 contains complete demographics.

Approximately 45% of respondents agreed or strongly agreed that NTPNs were able to practice competently with respect to eight competency statements (AACN, 2021). Table 2 summarizes these results. Notably, clinical nurses (i.e., frontline nurses, including charge nurses and preceptors) rated NTPNs' competence significantly more favorably than frontline leaders—Clinical Nurse Specialists (p = .01), Nurse Educators (p = .005), and Nurse Managers (p = <.001); there were no significant differences among frontline leaders or between frontline leaders and senior leaders (i.e., Directors and Chief Nurses). No significant differences were noted among care settings or regions of the country.

On average, respondents estimated their teams comprise 28% NTPNs; two-thirds of respondents noted a higher proportion of NTPNs post pandemic. Most agreed or strongly agreed that the increased proportion of NTPNs impacts quality and safety (56%), experienced nurses (67%), and nurse leaders (66%). Table 3 summarizes these results.

Table 2
New-to-Practice Nurses' Ability to Practice Competently

New-to-Practice Nurses' Ability to Practice Competently					
Survey Questions: Section 1	n (%)				
New-to-Practice Nurses Are Entering the Work Environment Prepared To	Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
Engage with patients to establish caring relationships[PCC 2.1] (demonstrate empathy[2.1a], compassion[2.1b], and mutual respect[2.1c]).	7 (1.9)	65 (17.2)	70 (18.5)	207 (54.8)	29 (7.7)
Communicate effectively with patients[PCC 2.2] (consider individual beliefs, values[2.2b], use various communication modes[2.2c], conduct sensitive conversations[2.2d], and demonstrate emotional intelligence[2.2f]).	9 (2.4)	110 (29.1)	100 (26.5)	142 (37.6)	17 (4.5)
Integrate their assessment skills in practice[PCC 2.3] (perform a clinically relevant assessment[2.3c], distinguish between normal and abnormal findings[2.3e], and communicate the findings of their assessment[2.3g]).	13 (3.4)	92 (24.4)	85 (22.5)	168 (44.6)	19 (5.0)
Diagnose actual or potential health problems and needs[PCC 2.4] (synthesize assessment data[2.4a], create a list of problems[2.4b], and prioritize problems[2.4c]).	21 (5.6)	114 (30.3)	122 (32.4)	107 (28.5)	12 (3.2)
Develop a plan of care[PCC 2.5] (apply results of screening in plans of care[2.4d], incorporate evidence-based interventions to improve outcomes and safety[2.5d], and anticipate outcomes of care[2.5e]).	13 (3.5)	116 (30.9)	105 (27.9)	127 (33.8)	15 (4.0)
Demonstrate accountability for care delivery[PCC 2.6] (implement plans of care[2.6a], delegate appropriately[2.6c], and monitor implementation of plans of care[2.6d]).	20 (5.3)	110 (29.3)	99 (26.3)	133 (35.4)	14 (3.7)
Perform effectively in teams[IP 6.2] (communicate professionally, accurately, and timely[6.1e], engage within the team as appropriate to one's scope of practice and competency[6.2c]).	11 (2.9)	72 (19.1)	96 (25.5)	178 (47.2)	20 (5.3)
Demonstrate professionalism[P9.1,P9.3] (reflect on one's actions and consequences[9.1b], demonstrate ethical behaviors[9.1c], change behaviors based on self and situational awareness[9.1d], take responsibility for one's actions[9.3d], and engage in professional activities[9.3e]).	22 (5.8)	86 (22.8)	92 (24.4)	156 (41.4)	21 (5.6)

Note. IPP, Interprofessional Partnership domain; P, professionalism domain; PCC, Patient-Centered Care domain.
Survey statements derived from AACN's *The Essentials: Entry-Level Professional Nursing Education Competencies* (2021).
Numeric footnotes designate subcompetencies.

Table 3
Composition of Nursing Teams and Associated Impacts

Composition of Nursing Teams and Associated Impacts Survey Questions: Section 2	*n* (%)		
	Lower Proportion of Novice Nurses	About the Same Proportion of Novice Nurses	Higher Proportion of Novice Nurses
How has the proportion of novice nurses (less than 18 months practice) changed since the pandemic?	24 (6.5)	96 (25.9)	250 (67.6)

Survey Questions: Section 3	*n* (%)				
	Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
The proportion of novice to experienced nurses in my unit/division/ organization is impacting care safety/quality.	19 (5.2)	50 (13.7)	92 (25.2)	161 (44.1)	43 (11.8)
The proportion of novice to experienced nurses in my unit/division/ organization is impacting experienced nurses.	14 (3.9)	42 (11.8)	62 (17.4)	177 (49.7)	61 (17.1)
The proportion of novice to experienced nurses in my unit/division/ organization is impacting nurse leaders.	9 (2.5)	39 (11.0)	74 (20.8)	178 (50.0)	56 (15.7)
Since the pandemic, nurses' sense of meaning and connection to their work has changed.	7 (2.0)	19 (5.3)	53 (14.8)	164 (45.9)	114 (31.9)

Qualitative Findings

One-hundred-eighty respondents (45%) provided open-ended responses to at least one question. From these responses, several themes were identified. Table 4 presents direct quotes illustrating each theme.

Experience Gained Over Time Prepares Nurses to Practice Competently

Respondents emphasized that both formal education and experience gained over time are important to NTPNs' ability to achieve competence. Many emphasized that learning to navigate the practice setting takes sufficient exposure but that the exposure gained through clinical rotations is often not enough to prepare nurses to

Table 4
Qualitative Themes

Qualitative Themes	
Theme	Quotes
Experience Gained Over Time Prepares Nurses to Practice Competently	"Competence comes with more experience." —311, Nurse Educator "[Competence] takes time but they can get there." —194, Nurse Educator "I do not believe that a lot of the 4-year colleges provide enough clinical time to expose new nurses to the hospital setting." —169, Nurse Manager "Those with experience with patients, outside of nursing school, perform better." —29, Nurse Educator "The… internship and externship programs are a huge success to help transition after graduation." —6, Clinical Nurse Specialist
Clinical Skills Gaps Are Prevalent	"Many will point out something that stood out to them during their assessment but will be unable to link it to the overall condition. [They] rely heavily on relaying this to the MD for interpretation." — 68, Clinical Nurse Specialist "New nurses seem to have knowledge of how to conduct focused assessments but not head-to-toe assessments. They cannot differentiate between normal and abnormal vital signs, lab data, and other relevant diagnostics." —118, Director "[They] have difficulty problem-solving…get stuck when there is not an established process. Not able to think about how to implement an intervention that is best for the patient." —129, Director "This feels new in that it started to develop during COVID and has persisted… we've lost the 'basics." —449, Director "We have so many new grads who have graduated with so few competencies to work in the hospital. I had a whole unit, 75% new grads, request education on a basic competency… Mobility is another one new grads think is the responsibility of the [physical therapist]. This is clearly in our scope of practice." —328, Director
Task Orientation Supercedes Critical Thinking	"Most novice nurses are just completing the task instead of using critical thinking to analyze patients' signs and symptoms, correlating with the monitor reading and lab results." —36, Clinical Nurse "[They] do not engage consistently with patients…[are] more task-driven based on identified tasks in the EMR."—256, Chief Nurse "It appears newer nurses are dependent on technology to prompt them, making them rely on a computer to identify gaps and safety points that are part of nurses' critical thinking. Computers should enhance and assist… nursing should not be depending on technology to assure they are critically thinking and doing their jobs." —184, Director

(*continued on next page*)

Table 4 (*continued*)

Qualitative Themes	
Theme	Quotes
Communication Skills Gaps Are Prevalent	"They lack confidence in having face-to-face conversations, be it with providers or navigating [patients'] complex emotional and behavioral issues." —526, Clinical Nurse Specialist "Many NTPNs have difficulty with skilled communication and difficult conversations." —80, Nurse Manager "The empathy and compassion are gone. The skill level is not high quality and the engagement is missing." —476, Nurse Manager "I have been seeing communication gaps… I have had to add a lot of emphasis on how to express empathy." —105, Nurse Manager "[They have] less feeling of connection to their patients." —342, Nurse Manager "Over the last four years, there has been a strong shift away from personal connections with patients. Everything is transactional and metric-based." —375, Nurse Educator "[They] often seem afraid to touch patients." —129, Director "If [they have] no experience, many seem afraid to go near the patients." —421, Nurse Manager "[They are] unable to effectively communicate with team members because they don't know their role." —68, Clinical Nurse Specialist "[They are] unsure where they fit in the care delivery team [and] do not seem to understand their role as care coordinators." —305, Director "Many are reluctant to communicate with attendings." —401, Nurse Manager "[They] have difficulties communicating assessments to providers." —92, Director "We are seeing challenges with communication between disciplines. Secure chat has taken over communication that used to be bedside, stunting relationship[*s*]… new grads are nervous to contact [providers]." —439, Director "In emergent clinical situations, oftentimes new nurses' first instincts are to turn to a secure chat device to send a message when what we really need is to pick up a phone and get the entire team to lay eyes on a patient." —526, Clinical Nurse Specialist "NTPNs can often work together appropriately but seem to struggle with anyone older or higher in the hierarchy, often lacking needed professionalism." —312, Nurse Educator "I observe many new nurses overly social on units… This is often a complaint of more experienced nurses." —539, Clinical Nurse Specialist
Generational Differences in Professional Behaviors	"Many NTPNs lack professionalism." —345, Nurse Manager "Professionalism is shifting with the new generation." —541, Director "[There is a] lack of accountability regarding mistakes and resistance to feedback." —76, Nurse Manager "[We] have seen an increase in the shared perspective of, 'no one showed me' or, 'no one explained this to me' as a response to feedback and in spite of proper training." —21, Nurse Manager "Newer nurses are not as accountable for their work. They leave tasks without telling the next nurse, and when they are given feedback or asked a question, there is little ownership." —74, Clinical Nurse Specialist "[We] continuously have to teach these nurses how to receive feedback without getting offended." —77, Nurse Manager "NTPNs see feedback as negative; [they] take it personally instead of constructively." —129, Director "I am seeing many young nurses unable to take constructive criticism in a healthy way to improve their professional practice. Often they will react in very unprofessional ways." —312, Nurse Educator "Phone/ air pod usage has risen at the bedside, requiring strict policies regarding personal phone use." —76, Nurse Manager "The younger generation is more likely to call out or need discussions about lateness than others." —385, Director "[NTPNs] seem to be less engaged in professional activities." —448, Nurse Educator "It is frustrating to teach the newer generation proper work etiquette." -—128, Nurse Manager "Nurse leaders spend significant time counselling novice nurses related to their interpersonal conflicts with colleagues, friends, and family, learning to communicate with different generations of nursing, and understanding themselves." —150, Clinical Nurse Specialist

(*continued on next page*)

Table 4 (*continued*)

Qualitative Themes	
Theme	Quotes
Stretched to Provide Support That Bridges Gaps and Maintains the Standard of Care	"Orientations are taking longer to complete…more nurses [are] on orientation simultaneously [and there is a] lack of skills and knowledge as a collective unit." —75, Nurse Educator "Many new nurses [are] being onboarded by novice preceptors (6 months – 2 years of experience), which affects their competency development and knowledge of policies and procedures." —103, Nurse Educator "New staff need constant supervision and support." —65, Clinical Nurse Specialist "[They] require more extensive support for longer periods of time off orientation. They tend to still struggle with becoming overwhelmed, prioritizing appropriately, communicating with the care team and leadership staff, and seeing the 'bigger picture' of patient care." —350, Clinical Nurse Specialist "There is a significant increase in clinical questions for charge nurses. Experienced nurses feel as if they are carrying their assignment and someone else's." —56, Clinical Nurse Specialist "Experienced nurses feel they need to keep a closer eye on novices because [they] sometimes fail to recognize patients' changing or deteriorating conditions." —97, Clinical Nurse Specialist "Novice nurses require a significant amount of one-to-one facetime with their leader. They need constant reassurance and recognition… and aren't shy about sharing about personal challenges, which creates additional strain." —541, Director "There is a huge burden and burnout on our experienced nurses when it comes to continued education, fixing mistakes, precepting, etc. It can be very exhausting having to continually worry about the work that other people are doing." —72, Nurse Manager "Experienced nurses report a sense of anxiety in the charge role as they are often the most experienced nurse." —67, Nurse Manager "The amount of responsibility placed on experienced nurses becomes overwhelming and is contributing to burnout… it is vital that as an organization we are mindful of this impact because it is going to cause issues in our ablity to retain experienced nurses." —357, Clinical Nurse Specialist "The experienced nurses are leaving because they are burnt out with having to do so much intensive precepting constantly." —328, Director "… the amount of time to justify a replacement, entering in the computer, reviewing resumes, coordinating interviews, conducting interviews, hiring, completing the orientation schedule, getting preceptors! It's exhausting." —124, Nurse Manager "[We are] working tirelessly to balance the skill mix in the schedule." —67, Nurse Manager "the constant churn of rehiring causes a lack of progress." —340, Nurse Educator "Many of the meeting agendas and safety concerns center around NTPNs instead of advancing nursing care." —62, Nurse Manager "There are many shifts, especially night shifts, staffed with a majority of novice nurses." —Nurse Educator, 315 "[We have seen] an uptick in medication errors." —81, Nurse Manager "[We see] a significant amount of 'missed care." —90, Clinical Nurse Specialist "Increased hospital-acquired pressure injuries year-to-year. CLABSIs were a 'never' event pre-pandemic and now we have 2–3 yearly." —317, Clinical Nurse Specialist "Clinical emergencies are more challenging to run effectively if there aren't more experienced RNs to step in and lead." —143, Clinical Nurse Specialist "[NTPNs are] unable to recognize and intervene for clinical deterioration." —272, Clinical Nurse Specialist "Novices don't have the experience to anticipate what might happen with a patient. We are seeing more near misses and rapid responses." —269, Nurse Educator
Loss of Connection and Engagement	"Seems like it's 'just a job'rather than having a deeper meaning." —88, Nurse Educator "It appears that the 'calling' and mission of nurses to their work and patients has shifted. More nurses, especially newer nurses, are transactional in their care." —172, Nurse Manager "There is no loyalty to the unit or organization. They are just looking for their next position. They don't want to join committees. They don't see the value in shared governance." —124, Nurse Manager "I often hear from experienced nurses that patients are sicker and the expectations are high with the same or less resources. They do not feel as valued as they once did." —73, Nurse Manager "Patients are more abrasive and aggressive, and nurses are forced to take the abuse… they do not want to engage with patients more than necessary." —363, Nurse Manager "Many staff feel the general population has become more acutely ill and less appreciative of the care [we] provide. This lack of respect has affected morale… they are feeling burnt out." —158, Nurse Manager
Increasing Education to Meet Learning Needs	"Added a second week to centralized orientation with additional time in charting and quality indicator skills… have safety day 6 weeks after start for demonstration of high risk skills and quality bundles." —327, Nurse Educator "[We need] shoulder-to-shoulder education." —389, Nurse Educator "Nurse Tech program that employs nursing students to work under a narrower scope of practice to get a head start on developing skills prior to graduation." —311, Nurse Educator

practice in the hospital. In contrast, respondents highlighted that NTPNs benefit from previous experience in other careers, as nurses' aides, or from externships.

Clinical Skills Gaps Are Prevalent

Respondents outlined various clinical skills gaps. They highlighted NTPNs' difficulties conducting assessments and contextualizing data to understand patients' clinical pictures. They noted NTPNs' challenges prioritizing care for multiple patients and applying assessment findings toward appropriate interventions. Lastly, several respondents raised concerns about NTPNs' basic nursing skills, noting gaps in competency and a lack of awareness around the nurses' role in delivering particular types of basic care (e.g., mobility) to patients.

Task Orientation Supercedes Critical Thinking

Exacerbating these gaps, many respondents noted increasing task orientation. They emphasized that NTPNs seem over-reliant on the electronic health record to direct care, prioritizing task completion and documentation over patients. Many suggested that nurses do what is prescribed in the electronic health record without a larger understanding of the patient's clinical picture, and therefore, are unable to intervene proactively or communicate therapeutically with patients.

Communication Skills Gaps Are Prevalent

Other noted skills gaps were relational, impeding NTPNs' connection with patients and team communication. Respondents noted NTPNs' lack of confidence with communication and, in particular, difficulty managing challenging conversations. Specific to patient interactions, many highlighted that NTPNs seem to lack empathy, engagement, and compassion. Several respondents even described NTPNs as afraid to talk with patients. Specific to team interactions, respondents described NTPNs as reluctant or struggling to communicate with attending physicians; many suggested these communication gaps stem from a lack of understanding about their role within the care team. They also pointed to specific challenges NTPNs have conveying assessment findings and a reliance on texting as a primary means of communication, which erodes team dynamics. Finally, respondents described strained communications with experienced team members, despite avid socialization with other NTPNs. This socializing was often described as unprofessional.

Generational Differences in Professional Behaviors

Compounding communication gaps, respondents highlighted a shift in professionalism among NTPNs. Many respondents outlined common unprofessional behaviors, including a lack of accountability, resistance and defensive reactions to feedback, and gaps in basic professional etiquette. Further, respondents highlighted that NTPNs seem to be less engaged in professional activities and self-driven development, which impedes learning and frustrates experienced team members.

Stretched to Provide Support That Bridges Gaps and Maintains the Standard of Care

Respondents described numerous traditional training methods used to support NTPNs' competency development, including preceptor-led orientation, charge nurse oversight, classroom education, and hands-on training. Respondents also highlighted, however, that many NTPNs require extended orientations, and a high volume are being oriented simultaneously. As a result, fewer experienced nurses are available to precept, and NTPNs are learning from novice peers. The collective level of knowledge and skill on units is diminished.

In turn, post orientation, NTPNs continue to require a great deal of emotional and tangible support to provide reassurance, prioritize care, communicate with the team, and understand patients' clinical pictures. Therefore, experienced nurses and leaders assume additional responsibilities to help NTPNs. They closely supervise NTPNs' care and answer numerous questions throughout each shift to support NTPNs in managing their patient assignments, developing their clinical skills, and maintaining the standard of care.

Many respondents described bolstering NTPNs as burdensome, contributing to fatigue, burnout, and turnover. Experienced nurses—who often have only a few years experience— feel overwhelmed and anxious assuming added responsibilities and worrying about others' patients. Their leaders worry they will leave due to burnout; in some organizations, leaders report they are already witnessing turnover. Formal leaders also described an increased workload related to the constant need to recruit, orient new staff, and design the schedule to balance skill mix safely; this churn is contributing to fatigue and frustration and preventing leaders from achieving progress with their teams. Instead, they are noticing adverse impacts to quality and safety. Several respondents emphasized increased risks on certain shifts. Others highlighted more medication errors, safety events, missed care and delays, and a diminished focus on nursing-sensitive indicators, contributing to complications.

Loss of Connection and Engagement

When asked about a sense of connection and meaning in their work, many respondents described a generational difference in NTPNs' connection to the profession. They emphasized that NTPNs approach their work as a "just a job" without a sense of mission or loyalty to their organization or unit and that they are reticent to engage in professional activities. This loss of connection was not limited to NTPNs, however. Respondents described a lingering trauma post pandemic and feeling undervalued by their organizations and the public. This feeling combined with increased demands—sicker and more aggressive patients, more tasks placed on nurses, including covering roles that were cut or made remote, tighter staffing, and a significant training burden—contributes to low morale, apathy, and disconnection. As a result, nurses are more focused on protecting their work-life balance and seeking more flexible work.

Increasing Education to Meet Learning Needs

When asked about strategies used to support NTPNs' development, respondents commonly suggested education. Many mentioned organization-wide efforts like orientation, residency programs, skills days, simulation, or preceptor development. Respondents also noted unit-level efforts, including specialty training, extending orientation, mentoring programs, frequent leader–preceptor–orientee check-ins, and providing clinical or emotional support. Many highlighted the importance of educators working at the organization, population, and/or unit levels to develop or revise content and provide support at the bedside. Some pointed to the creation of new roles and a few suggested novel training models. Numerous respondents, however, reported insufficient educational resources.

Discussion

In 2021, AACN released *The Essentials*, calling for a shift toward CBE. CBE ties academic coursework to professional expectations, focusing on "what learners can do with what they know" rather than what they are taught; the goal is to prepare "practice-ready nurses" who have skills to transition into the workforce and provide care in a complex and evolving healthcare environment (American Association of Colleges of Nursing, n.d.; National Academies of Sciences, 2016). To implement CBE, AACN calls on schools of nursing and practice organizations to partner (American Association of Colleges of Nursing, n.d.).

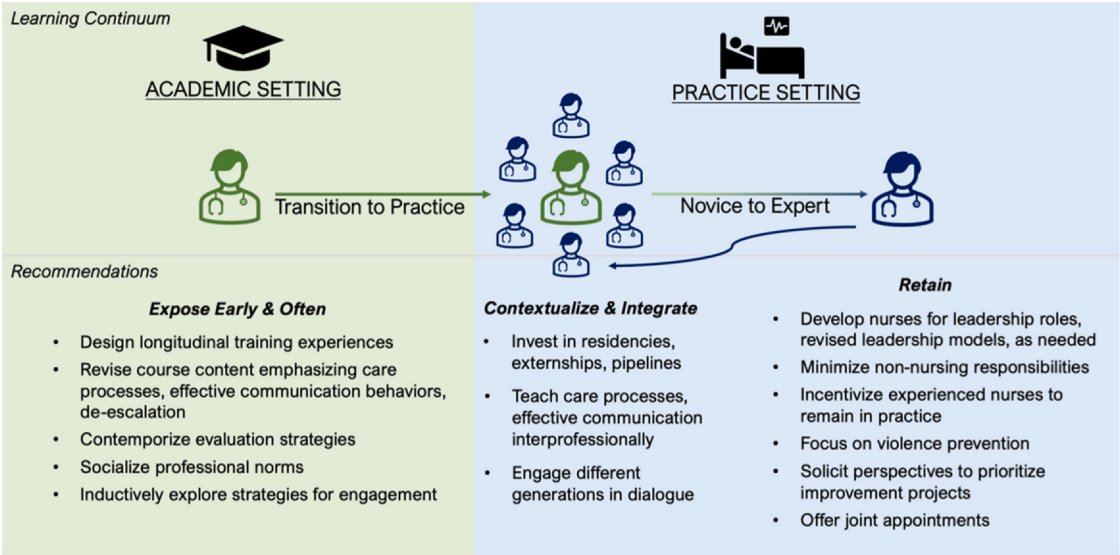

Figure 2. Strategies to engage nurse learners along the lifelong learning continuum.

We agree that academic-service partnership—with support from policymakers—is essential for designing and implementing strategies to advance nursing education and practice at various points along the learning continuum, and in turn, fulfill our obligation to the public. Our study strengthens this effort by offering a view of the current state from the perspective of one group of key informants—practice leaders. Their perspectives emphasize several key findings. We recommend these findings be incorporated into shared efforts to implement CBE and fortify the nursing workforce, focusing broadly on (a) exposing NTPNs to clinical experiences early and often, (b) contextualizing care and integrating NTPNs into communities of practice, and (c) retaining experienced nurses. See Figure 2.

Key Finding 1: Building Competence Takes Time and Is Achieved Through Experience

Recommendation 1 Expand opportunities for NTPNs to gain cumulative exposure to clinical settings to better understand the care context.

1a. Practice leaders should invest in externship, career pipeline, and nurse residency programs that provide clinical and classroom training pre- and post hire (Swan et al., 2024).

1b. Practice and academic leaders should partner to develop novel training opportunities, like longitudinal experiences where nurses gain repeated exposure and cumulative feedback within a given setting (Goers et al., 2024).

1c. Policymakers should allocate funding toward workforce initiatives aimed at designing and evaluating novel training models.

Key Finding 2: NTPNs Exhibit Clinical and Communication Skills Gaps and Increased Task Orientation

Recommendation 2 Update teaching and evaluation strategies used pre- and post licensure to address noted gaps.

2a. When teaching clinical skills, avoid decontextualization. Academic and practice leaders should emphasize processes of care delivery, incorporating a focus on coordinating and executing care within a real-world context. Teaching methods should prompt learners to consider how the nurses' role interfaces interprofessionally, what clinical changes should be anticipated, how concerns would be escalated, and how communication might unfold with patients and families. Training should be interprofessional, when possible (Curley et al., 2024).

2b. Academic and practice leaders should incorporate communication-specific content into training, focusing on behaviors that convey empathy.

2b. Academic leaders should partner with practice leaders to contemporize student evaluations, ensuring feedback promotes reflection about soft skills. For example, clinical evaluations could incorporate criteria from employee evaluations or be cumulative between semesters. Students could be asked to solicit feedback from nurse preceptors or patients.

Key Finding 3: Generational Differences in Communication and Professional Behaviors Create Team Tension That Can Impede Shared Learning

Recommendation 3 Leverage generational diversity and limit stereotyping that creates divisiveness.

3a. Academic leaders should ensure curricular revisions emphasize and evaluate students' professional behaviors, including giving and receiving feedback. Practice leaders should ensure new employees are onboarded with clear expectations applied consistently for all staff.

3b. Practice and academic leaders should create opportunities for dialog among different generations of nurses, emphasizing humility and curiosity, getting to know one another, and exploring each other's strengths (Pearce, 2024).

3c. Academic and practice leaders should use inductive methods to explore generational perspectives around meaning in work and how best to receive feedback, communication, and education. They should engage nurses of varied generations in designing communication strategies.

Key Finding 4: Experienced Nurses and Nurse Leaders Absorb a Greater Training Effort—Extending Orientation and Providing More Post-Orientation Support—Than in Pre-Pandemic Years, Which Alongside Operational Demands, Leads to Frustration, Fatigue, Burnout, and Turnover

Recommendation 4 Provide meaningful support to experienced nurses and nurse leaders.

4a. Practice leaders should provide development for nurses assuming leadership roles that address necessary leadership competencies, leverage peer networks, and target emerging and current leaders (American Organization of Nurse Leaders, 2024).

4b. Practice leaders should work with interdisciplinary colleagues to minimize the non-nursing responsibilities nurses assume.

4c. Practice leaders should survey current training efforts to identify redundancies and best practices. Subsequently, scale best practices and eliminate unnecessary duplication. Consider whether leaders' role expectations match their expertise or if revised leadership models are needed (Curley et al., 2024).

4d. Well-rounded approaches to mitigating workplace violence are essential. Practice leaders should implement violence prevention programs. Academic leaders should include course content on de-escalation. Both should advocate for healthcare worker protections. Policymakers should enact legislation penalizing violence toward healthcare workers and allocate funding to develop prevention strategies.

Key Finding 5: Eroded Communities of Practice Contribute to Quality and Safety Concerns

Recommendation 5 Retain and leverage experienced staff.

5a. Practice leaders should incentivize experienced nurses to remain in practice, for example, by creating roles or work assignments like virtual mentoring programs, that offer flexibility while keeping expertise at the bedside (Tibbe et al., 2023).

5b. Practice leaders should employ data-informed staffing models to match nurses' competencies with patient assignments (Curley et al., 2024).

5c. Practice and academic leaders should solicit feedback from experienced staff regarding areas of greatest quality and safety risk. These areas should be prioritized for improvement efforts, including graduate student projects.

5d. Academic leaders should offer experienced nurses joint appointments. These appointments could build reciprocal academic partnerships to help practice leaders gain greater insight into the academic setting.

5e. Policymakers should allocate funding to help practice organizations and schools share expert nurses via joint appointments.

Conclusion

Health care continues to grow more complex. Nurses are caring for sicker patients using a wide array of technology (Advisory Board, 2019). At the same time, the pandemic catalyzed experienced nurses' departure from the workforce (Martin et al., 2023), disrupting nursing communities of practice that have, for years, been essential to cultivating nursing competency and expertise. Experienced nurses and practice leaders are stretching to bridge the gaps that have emerged. Yet, expecting individuals or single organizations to solve these issues in isolation risks further attrition of experienced nurses and leaders and a greater loss of expertise at the bedside. To halt this trend and ensure the nursing workforce is prepared to meet patient care demands, practice, academic, and policy leaders must partner to implement strategies that bolster communities of practice from where they begin—in nursing school—to where they are sustained—in the practice setting.

Funding

There was no funding for this work.

CRediT Statement

Aditi D. Rao: Writing – review and editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. *Evan P. Anderson*: Writing – review and editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. *Beth A. Smith*: Writing – review and editing, Writing – original draft, Conceptualization. *Matthew McHugh*: Writing – review and editing, Writing – original draft, Conceptualization. *Regina S. Cunningham*: Writing – review and editing, Writing – original draft, Conceptualization.

Declaration of Competing Interest

The authors declare no conflicts of interest.

References

AACN. (2021). The essentials: Core competencies for professional nursing education.

Advisory Board. (2019). *The experience-complexity gap*. https://www.advisory.com/topics/nurse-staffing-and-productivity/2019/03/nursing-shortage.

American Association of Colleges of Nursing. (n.d.). *A guide to curricular transitions for competency-based education*. Retrieved October 12, 2024, from https://www.aacnnursing.org/Portals/0/PDFs/Essentials/Guide-to-Curricular-Transition-for-Competency-Based-Education.pdf.

American Organization of Nurse Leaders. (2024). *Trends and Innovations in Nurse Manager Retention Fall 2024 Report*. https://www.aonl.org/Quantifying-Nurse-Manager-Impact.

Auerbach, D. I., Buerhaus, P. I., Donelan, K., & Staiger, D. O. (2024). Projecting the future registered nurseworkforce after the COVID-19 pandemic. *JAMA Health Forum, 5*(2), Article E235389. https://doi.org/10.1001/jamahealthforum.2023.5389

Baartman, L. K. J., & De Bruijn, E. (2011). Integrating knowledge, skills and attitudes: Conceptualising learning processes towards vocational competence. *Educational Research Review, 6*(2), 125–134. https://doi.org/10.1016/j.edurev.2011.03.001

Benner, P. (1984). *From novice to expert: Excellence and power in clinical nursing practice*. Addison-Wesley.

Billett, S. (2004). Workplace participatory practices: Conceptualising workplaces as learning environments. *Journal of Workplace Learning, 16*(6), 312–324. https://doi.org/10.1108/13665620410550295

Curley, M. A. Q., Zalon, M. L., Seckel, M. A., Alexandrov, A. W., Sorce, L. R., Kalvas, L. B., Hooper, V. D., Balas, M. C., Vollman, K. M., Carr, D. S., Good, V. S., Latham, C. L., Carrington, J. M., Hardin, S. R., & Odom-Forren, J. (2024). Call to action: Blueprint for change in acute and critical care nursing. *Nursing Outlook, 72*(6), Article 102271. https://doi.org/10.1016/j.outlook.2024.102271

Elo, S., & Kyngäs, H. (2008). The qualitative content analysis process. *Journal of Advanced Nursing, 62*(1), 107–115. https://doi.org/10.1111/J.1365-2648.2007.04569.X

Goers, J., Humphrey, K., Oja, K., & Mulkey, D. (2024). Bridging the experience-complexity gap with longitudinal clinical placements. *Nursing Outlook, 72*(5), Article 102221. https://doi.org/10.1016/j.outlook.2024.102221

Graneheim, U. H., & Lundman, B. (2004). Qualitative content analysis in nursing research: Concepts, procedures and measures to achieve trustworthiness. *Nurse Education Today, 24*(2), 105–112. https://doi.org/10.1016/j.nedt.2003.10.001

Hill, K. S. (2010). Improving quality and patient safety by retaining nursing expertise. *Online Journal of Issues in Nursing, 15*(3), https://doi.org/10.3912/OJIN.VOL15NO03PPT03

Hsieh, H., & Shannon, S. E. (2005). Three approaches to qualitative content analysis. *Qualitative Health Research, 15*(9), 1277–1288.

Kavanagh, J. M., & Sharpnack, P. A. (2021). Crisis in competency: A defining moment in nursing education. *Online Journal of Issues in Nursing, 26*(1), https://doi.org/10.3912/OJIN.VOL26NO01MAN02

Kavanagh, J. M., & Szweda, C. (2017). A crisis in competency: The strategic and ethical imperative to assessing new graduate nurses' clinical reasoning. *Nursing Education Perspectives, 38*(2), 57–62. https://doi.org/10.1097/01.NEP.0000000000000112

Korstjens, I., & Moser, A. (2018). Series: Practical guidance to qualitative research. Part 4: Trustworthiness and publishing. *European Journal of General Practice, 24*(1), 120–124. https://doi.org/10.1080/13814788.2017.1375092

Lincoln, Y. S., & Guba, E. G. (1985). *Naturalistic inquiry*. SAGE.

Martin, B., Kaminski-Ozturk, N., O'Hara, C., & Smiley, R. (2023). Examining the impact of the COVID-19 pandemic on burnout and stress among U.S. nurses. *Journal of Nursing Regulation, 14*(1), 4. https://doi.org/10.1016/S2155-8256(23)00063-7

Mchugh, M. D., & Lake, E. T. (2011). Understanding clinical expertise: Nurse education, experience, and the hospital context. *Research in Nursing & Health, 33*(4), 276–287. https://doi.org/10.1002/nur.20388

National Academics of Sciences, Engineering, and Medicine. (2016). Competency Based Accreditation and Collaboration. In Exploring the Role of Accreditation in Enhancing Quality and Innovation in Health Professions Education: Proceedings of a Workshop. National Academies Press (US).

Pearce, N. (2024). Leading the 6-Generation Workforce.

Swan, B. A., Jones, K. D., Hayes, R., Kaligotla, L., McDermott, C., Rodriguez, J., & McCauley, L. (2024). Reject the "Practice Readiness Myth": Ask if systems are ready for nursing graduates instead. *Nursing Outlook, 72*(5), Article 102181. https://doi.org/10.1016/j.outlook.2024.102181

Tibbe, M., Arneson, S., & Welsh, C. (2023). Rise of the virtual nurse. *AACN Advanced Critical Care, 34*(4), 314–323. https://doi.org/10.4037/aacnacc2023391

Tynjälä, P. (2008). Perspectives into learning at the workplace. *Educational Research Review, 3*(2), 130–154. https://doi.org/10.1016/j.edurev.2007.12.001

Wenger, E. (1998). Communities of practice: Learning as a social system.c